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Hongli Li, Charles Vincent Hunter & Jacquelyn Anne Bialo

To cite this article: Hongli Li, Charles Vincent Hunter & Jacquelyn Anne Bialo (2021): A Revisit of Zumbo's Third Generation DIF: How Are We Doing in Language Testing?, Language Assessment Quarterly, DOI: [10.1080/15434303.2021.1963253](https://doi.org/10.1080/15434303.2021.1963253)

To link to this article: <https://doi.org/10.1080/15434303.2021.1963253>



Published online: 30 Sep 2021.



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A Revisit of Zumbo's Third Generation DIF: How Are We Doing in Language Testing?

Abstract

The purpose of this study is to review the status of differential item functioning (DIF) research in language testing, particularly as it relates to the investigation of sources (or causes) of DIF, which is a defining characteristic of the third generation DIF (Zumbo, 2007a). This review included 110 DIF studies of language tests dated from 1985 to 2019. We found that DIF researchers did not address sources of DIF more frequently in recent years than in earlier years. Nevertheless, DIF research in language testing has expanded with new DIF analysis procedures, more grouping variables, and more diversified methods for investigating sources of DIF. In addition, in the early years of DIF research, methods to identify sources of DIF relied heavily on content analysis. This review showed that while more sophisticated statistical procedures have been adopted in recent years to address sources of DIF, understanding sources of DIF still remains a challenging task. We also discuss the pros and cons of existing methods to detect sources of DIF and implications for future investigations.

Keywords: Third Generation Differential Item Functioning, Review, Language Testing

摘要

差别试题功能（DIF）用来检测能力相匹配的群体在应答某个题目时是否表现出差异。本文旨在对语言测试领域的 DIF 研究进行一个综述，尤其是引起 DIF 的来源是什么。DIF 来源是 Zumbo 的第三代 DIF 研究的一个主要特点。本综述包括了 1985 年到 2019 年之间的 110 篇语言测试领域的 DIF 文章。和以往相比，近些年的语言测试领域的 DIF 研究并没有更加频繁地研究 DIF 的来源。我们发现 DIF 研究扩展到了更新的 DIF 分析方法，更多的 DIF 分组变量，和更加多元化的研究 DIF 来源的方法。在早期关于 DIF 来源的文章中，研究人员使用的方法主要是试题内容分析。近些年来，研究人员开始使用更加复杂的统计分析方法。然而了解 DIF 的来源仍然是一个充满挑战性的题目。最后，我们讨论了探寻 DIF 来源的各个分析手段的优缺点，也对未来研究方法提出了建议。

Introduction

Differential item functioning (DIF) methods evolved as a response to concerns about test fairness (Kunnan, 2007). According to the *Standards for Educational and Psychological Testing*, DIF occurs when “different groups of test takers with similar overall ability, or similar status on an appropriate criterion, have, on average, systematically different responses to a particular item” (AERA, APA, & NCME, 2014, p.16). However, DIF is distinct from item bias as items showing DIF are not necessarily biased. Item bias occurs when examinees in one group are less likely to answer an item correctly than examinees in another group because of some characteristics of the test item or testing situation that are irrelevant to the purpose of the test (Zumbo, 2007a; Zumbo et al., 2015). Therefore, DIF is regarded as a necessary but not sufficient condition for item bias (Zumbo, 1999).

While many methods have been proposed to investigate DIF and numerous DIF studies have been conducted, it remains a challenging task to understand why DIF occurs. To that end, in his seminal article, Zumbo (2007a) reviewed the history of DIF and described three generations of DIF praxis and theorizing. The first generation of DIF was characterized by concerns regarding item bias. Many early methods compared two groups of examinees, with the *focal group* referring to the minority group and the *reference group* referring to the majority group. First generation DIF also saw a shift in terminology from *item bias* to *differential item functioning*. The second generation of DIF separated *impact* (i.e., true difference) from *bias* and introduced sophisticated statistical methods to investigate DIF items. Finally, Zumbo (2007a) regarded the inquiry of understanding why DIF occurs as a sign of the third generation of DIF. Different from the first and second generation, third generation DIF saw sources of DIF move beyond the item characteristics that were traditionally used to explain DIF to also include “test

situations” as potential sources of DIF. Furthermore, by drawing on contextual variables to explain DIF, third-generation DIF is aligned with an explanation-focused view of test validity that includes contextual sources to account for variation in item responses (Zumbo, 2007b; Zumbo, 2009).

Over 13 years have passed since the publication of Zumbo’s article; however, it remains unknown how much DIF researchers have continued to embody the third generation praxis of addressing sources of DIF (Zumbo, 2007a). Therefore, the purpose of this study is to review the status of DIF research, primarily as it relates to the methods used to investigate sources of DIF. In particular, we focused on whether researchers looked at sources of DIF and how those sources were investigated. DIF research in the field of language testing has been very prolific and enabled us to conduct a robust review of DIF strategies over the last several decades. As such, we restricted our review to DIF studies in language testing only. Three research questions are addressed: 1) What are the characteristics of DIF studies in language testing? 2) How often do DIF studies in language testing investigate sources of DIF? 3) What methods have been used to investigate sources of DIF in language testing?

Literature Review

Over the past 30 years or so, numerous methods to detect DIF have been developed and used across a large number of studies (see Mapuranga et al., 2008; Penfield & Lam, 2005; Zwick, 2012), including delta-plots (Angoff, 1972), the standardization procedure (Dorans & Kulick, 1986), Mantel–Haenszel technique (MH; Holland & Thayer, 1988), logistic regression (Swaminathan & Rogers, 1990), item response theory (IRT)–likelihood ratio tests (Steinberg et al., 1990), and Simultaneous Item Bias Test (SIBTEST; Shealy & Stout, 1993). Zumbo et al.

(2015) classified the various DIF methods into four frameworks: (a) item response modeling via contingency tables and/or regression models, (b) IRT, (c) multidimensional models, and (d) latent class methods.

The first framework includes methods such as the MH or logistic regression, which have been very widely used in the last few decades. The major difference between the first framework and the second framework (i.e., the IRT approach) is how the groups are matched. While the MH and logistic regression methods match examinees based on their total ability, the IRT approach does not condition on total ability because it is believed that the ability distribution has been “integrated out” (Zumbo, 2007a). With the IRT approach, DIF is determined by calculating the area between item response functions or comparing the IRT item parameters between groups. According to Zumbo et al. (2015), the first and the second frameworks are purely statistical approaches. If researchers using those methods are interested in investigating sources of DIF, they usually have to move beyond statistical procedures by, for example, inviting content experts to review test items either before or after DIF analysis or asking examinees for their perceptions about test items.

The third framework of “the multidimensional models” is different from the first two frameworks in that the investigation of sources of DIF is directly incorporated into the analysis. In this framework, DIF is believed to result from multidimensionality because one or more secondary dimensions may affect examinee performance differently (Roussos & Stout, 1996; Shealy & Stout, 1993). SIBTEST is one multidimensional-based approach to DIF in which researchers start by using substantive evidence (e.g., expert content review, student think-aloud protocols) to make hypotheses about a secondary dimension causing DIF (Jang & Roussos, 2009; Roussos & Stout, 1996; Shealy & Stout, 1993). Secondary dimensions are further labeled as

auxiliary or nuisance dimensions. Auxiliary dimensions reflect intentional differences that are regarded as “benign”, because DIF caused by auxiliary dimensions does not lead to test unfairness. In contrast, unintended differences caused by nuisance dimensions that are unrelated to the intended test construct are regarded as “adverse” (Roussos & Stout, 1996). After the SIBTEST analysis, researchers interpret the results and explain the potential sources of DIF. In terms of investigating sources of DIF, however, SIBTEST only considers item characteristics but not the context of test administration or lives of the examinees. Therefore, SIBTEST is considered as a second-generation method because it does not consider testing situation as a possible reason for DIF (Zumbo, 2007a; Zumbo et al., 2015).

The fourth framework is the latent class approach, which is categorized as a third generation DIF method (Zumbo et al. 2015). Traditional DIF methods typically use observed grouping variables, such as gender, language of the assessment, or ethnicity. The assumption is that the observed groups are homogeneous; however, some research indicates that the observed groups are, in fact, heterogeneous and that a given source of DIF may not apply equally to everybody in the group (Cohen & Bolt, 2005; De Ayala et al., 2002). The latent class approach addresses the question of whether there is a potentially important grouping variable that is not directly observed. Many DIF researchers have adopted this method and tested their hypothesized source of DIF using latent variable mixture models, including mixture IRT and mixture Rasch methods to detect the latent groups (e.g., Cohen & Bolt, 2005; De Ayala et al., 2002; von Davier & Yamamoto 2007; Webb, et al., 2008).

Furthermore, Zumbo et al. (2015) introduced an ecological model of item responding as a third generation DIF method. With the ecological model, researchers not only use psychocognitive factors but also sociological, structural, community, and contextual variables to explain

item response patterns and DIF (Zumbo & Gelin, 2005). By thinking beyond the test structure itself to also include the “testing situation” as a possible reason for DIF, DIF researchers moved past the multidimensional model of second generation DIF (Zumbo et al., 2015). Such an explanation-focused view of DIF has been informed by the explanation-focused view of test validity (Zumbo, 2007b). The ecological model is highly versatile and can be the basis for many statistical and psychometric methods of DIF analysis (Zumbo et al., 2005).

Zumbo et al. (2015) demonstrated how the ecological model of item responding can be incorporated into a latent class approach to DIF with data from the Programme for International Student Assessment (PISA). In the first step of the approach by Zumbo et al. (2015), traditional logistic regression is used to identify DIF. The second step is to decide on the number of latent classes. In the third and fourth steps, predictors of latent class membership are included to examine how contextual variables may help understand the moderated functioning of DIF (Zumbo & Gelin, 2005). Within an ecological model of item responding, Zumbo et al. (2015) showed levels of the explanatory variables they examined, such as personal characteristics, teacher, classroom, school context, family and ecology outside of school, and characteristics of the community, neighborhood, state, and nation. Woitschach et al. (2019) applied the ecological model of item responding and combined it with a multilevel regression analysis. Taking into account school and country characteristics, they found about 32% of the science items on the Third Regional Comparative and Explanatory Study (TERCE) showed gender DIF. The authors concluded that the main source of DIF was related to the human developmental index of the 15 participating countries, which is a composite index of life expectancy, education, and average income.

In summary, DIF analysis is an important strategy used to address test fairness, which is a significant topic in educational measurement. While many DIF methods have been proposed and numerous DIF studies have been conducted, understand why DIF occurs remains a challenging task. Previously, Ferne and Rupp (2007) synthesized DIF research in language testing between 1990 and 2005, and Li et al. (2013) reviewed gender-related DIF in reading tests up to 2012. However, while both reviews provided general pictures of DIF studies, neither specifically touched on the issue of sources of DIF. As such, in this study, we review the status of DIF analysis in language testing, particularly as it relates to the third-generation praxis of investigating sources of DIF.

Methods

Searching Studies

Studies that met the following criteria were included in our review. First, only DIF studies with language tests were included. For the purpose of this review, we adopted the definition of “language tests” used by Ferne and Rupp (2007) in their DIF synthesis study: “Language tests seek to make inferences about the communicative ability of learners in their native or nonnative languages” (p. 118). For example, Farrington et al. (2015) conducted DIF analysis with Revised Get Ready to Read! (GRTR-R), which is a screening measure to assess preschool children’s print knowledge and phonological awareness. Both skills fall under the purview of language skills, so we regarded Farrington et al. as eligible for inclusion. We included studies that involved multiple content areas as long as the DIF results related to language skills were presented separately. Second, DIF techniques had to be applied to a real dataset; any studies using only simulated data were not included. Third, we focused on studies

that used DIF to evaluate test functioning. As such, we excluded studies where DIF methods were being used as a statistical technique for other purposes. For example, Li et al. (2017) was not included because they used a DIF method to examine the instructional sensitivity of a test. Finally, only peer-reviewed journal articles published in English before 2020 were considered, and there was no limit in regard to the starting year.

To begin with, we searched both ERIC and PsycINFO using keywords “differential item functioning” or “DIF” in the whole document, combined with the following keywords one at a time: Language, Linguistic, Verbal, Literacy, Reading, Writing, Speaking, Listening, Grammar, Vocabulary, Spelling, Syntactic, Discourse, Pragmatic, Lexical, Communicative, etc. We also manually searched issues of *Language Testing*, *Language Assessment Quarterly*, and *Language Testing in Asia* as these are the (primary) peer-reviewed journals in language testing. Finally, we searched the reference lists of previous DIF review studies in language testing, such as Ferne and Rupp (2007) and Li et al. (2013). The first author and the second author read each result to determine if the article met the inclusion criteria. As a result, 110 studies met our inclusion criteria (see references for the list of studies).

Coding Studies

We first drafted a coding sheet based on existing literature and prior DIF reviews (e.g., Ferne & Rupp, 2007; Li et al., 2013), with an emphasis on coding whether sources of DIF were examined in the study. After each author coded a small proportion of the studies using the coding sheet, the codes were finalized based on several rounds of discussions and revisions. As the focus of this review is to understand whether DIF studies in language testing have embodied the “third generation DIF” (i.e., understanding why DIF occurs), we separated the 110 studies into three categories based on whether and how the researchers investigated sources of DIF. In

Category A, the authors did not consider the source of DIF at all in their research (e.g., Chen & Fu, 2018; Keuning & Verhoeven, 2007; Sasaki, 1991). In Category B, the authors mentioned the source of DIF in their work, but did not collect systematic empirical evidence or conduct statistical analyses to examine the source of DIF. Many times, authors of Category B studies provided plausible interpretations of why DIF might have occurred based on their literature review or ad hoc observations (e.g., French & Gotch, 2013; Stubbe, 2011). Such explanations were generally speculative due to the lack of empirical evidence. In Category C, the authors systematically investigated sources of DIF with empirical evidence and/or statistical analyses. For example, Category C authors might have invited content experts to review and code item characteristics and then correlate those characteristics with the presence and/or magnitude of DIF (e.g., Freedle & Kostin, 1997; Ryan & Bachman, 1992). Sometimes, DIF was studied using a multidimensional approach, such as SIBTEST, in which nuisance dimensions thought to cause DIF were hypothesized *a priori* and then tested (e.g., Gierl, 2005; Jang & Roussos, 2009). In more recent years, some researchers also controlled for a series of contextual variables in order to examine whether including those variables may have contributed to the presence and/or magnitude of DIF (e.g., Sandilos et al., 2015). Similarly, other studies also used the ecological model of item responding to explain DIF with different layers of explanatory variables (e.g., Zumbo et al., 2015).

In addition to coding approaches to investigating DIF sources, we recorded other characteristics of the studies. First, we coded the year that the study was published and the geographic location of where the data were collected in each study. Second, we coded the journal where it was published. We further categorized the journals into three categories: journals focused on language testing (e.g., *Language Testing*, *Language Assessment Quarterly*, *Language*

Testing in Asia), journals focused on educational measurement (e.g., *Journal of Educational Measurement*, *International Journal of Testing*, *Applied Psychological Measurement*, *Educational and Psychological Measurement*, *Psychometrika*, *Educational Assessment*, *Applied Measurement in Education*), and journals from any other disciplines that did not fall into either of those categories (e.g., *Journal of Cross-Cultural Psychology*, *Journal of Language Disabilities*, *Reading and Writing*). Then we coded the content of the test being examined in each study, the language of the test, and the test takers' native languages (or languages used at home). Finally, we coded DIF grouping variables (i.e., reference group and focal group), DIF detection methods, and if any DIF items were flagged. After some initial training and calibration of rater reliability, the second author coded all the studies, and the first author checked the coding of each study multiple times. Any discrepancies were resolved through discussion and negotiation until a consensus was reached.

Data Analysis

To answer research question 1 ("What are the characteristics of DIF studies in language testing?"), we calculated the frequency and percentage of DIF studies across different characteristics. To answer research question 2 ("How often do DIF studies in language testing investigate sources of DIF?"), we presented the frequency of DIF study characteristics across our three categories of DIF source detection practices (Category A = did not mention the source of DIF, Category B = briefly mentioned the source of DIF, Category C = systematically examined the source of DIF). We also conducted Chi-square association tests to examine whether the relationships between DIF study characteristics and DIF detection practices were or were not significant at the .05 level. Due to the small frequency in some cells in Table 2, we only included categories with relatively large sample sizes for the Chi-square association test. To answer

research question 3 (“What methods have been used to investigate sources of DIF in language testing?”), we summarized our findings across three time periods: before 2000, 2000 to 2009, and 2010 to 2019.

Results

Research Question 1: What are the Characteristics of DIF Studies in Language Testing?

As a result of the systematic search, we included 110 studies in this review. Table 1 displays characteristics of: a) the studies themselves, such as time period of the publication date, publication journal, geographic area where data were collected; b) the tests being examined (e.g., test content, language); c) test-takers (e.g., native language); and d) DIF variables (e.g., presence of DIF items, DIF grouping variable, and DIF method). The earliest DIF study in language testing was by Chen and Henning in 1985, who examined the source of DIF using content analysis. As seen in Figure 1, the number of DIF studies in language testing increased after 2002 and peaked in 2014 with 12 studies published that year, followed by nine studies in 2012 and 2015, and eight studies in 2007 and 2011. The number of DIF studies lowered a bit towards the end of the 2010s, with seven studies in 2018 and three studies in 2019. We also divided the year of publication into three time periods. As shown in Table 1, there were 15 studies before 2000, 33 studies from 2000 to 2009, and 62 studies from 2010 to 2019.

Insert Figure 1 here

Insert Table 1 here

Data were collected from the U.S. or Canada in 49 studies, followed by 25 studies from Asian countries, 13 studies from European countries, six from Australia or New Zealand, and four from other regions (Guatemala, Nigeria, Saudi Arabia, and South Africa). Thirteen studies

included data from multiple countries, with those studies typically involving large-scale tests such as the SAT or PISA. A total of 27 studies were published in language testing journals, 41 studies were published in educational measurement journals, and 42 studies came from other journals, usually in the areas of educational research or language studies.

We also examined different aspects of the tests that were examined for DIF. For example, regarding the test content area, 41 studies focused on only reading comprehension tests, 12 studies focused on only vocabulary tests, and seven studies focused on only listening tests. In as many as 46 studies, the test being assessed for DIF measured multiple language skills. For example, many researchers examined DIF in the SAT verbal section, which simultaneously evaluates several language skills. We further coded the test language. Because some tests were offered in multiple languages, we counted the frequencies of languages that occurred in the 110 studies. The most frequent test language was English, which was used in 83 studies, followed by Spanish and French, Russian, German, Chinese, and Arabic.

In addition to characteristics of the tests being examined for DIF, we looked at variables related to test takers and whether they were English as a Second Language (ESL) learners. Specifically, for the 83 studies that used English language tests, we coded test takers' native language. While about half of the studies included test takers who spoke English only (33.7%) or English plus another language (18.1%), the rest of the studies included individuals who spoke a language other than English. The most frequently reported native languages were Farsi and Korean, followed by Japanese and Turkish. In 18 studies, students were from multiple countries or areas, and several native language groups were involved in an analysis, such as Chinese, Arabic, Thai, Indonesian, Portuguese, Hindi, Polish, Greek, etc. Sometimes, researchers moved beyond native languages and evaluated different dialects. For example, in Stubbe (2011), all

test-takers spoke German, but because they were from three different countries, the Progress in International Reading Literacy Study (PIRLS) reading assessments were translated from English into German dialects for Austria, Germany, and Luxembourg. The author found that differing translations were responsible for some DIF items between countries.

Table 1 also shows DIF-related characteristics. Among the 110 studies eligible for our review, DIF items were identified in 103 of them, with only seven studies not finding any DIF items. However, the criteria authors used to determine DIF varied widely across studies, and depended on factors such as the statistical rules authors used to flag items, considerations of practical significance in addition to statistical significance, sample sizes, type I errors, etc. As such, in order to synthesize results from studies with different DIF criteria, we adopted the conclusion provided by the author(s) regarding whether DIF items were found in their study. Our finding that most studies detected DIF items is consistent with previous research. For example, in their review of gender DIF in reading tests, Li et al. (2013) found that DIF items were present in 16 out of 18 DIF studies.

In Table 1, we also report the frequency of grouping variables for each DIF study. Many authors used multiple grouping variables in one single DIF study. Gender was the most frequently used grouping variable, followed by language (such as native language, language used at home, mother's native language, language of the test, etc.), ethnicity, and age. This finding is aligned with the pattern found in Ferne and Rupp's (2007) review of DIF studies in language testing. However, the range of "other" DIF grouping variables, such as socioeconomic status (e.g., Heppt et al., 2015), academic discipline (e.g., Liu, 2011; Pae, 2004a), urban vs. rural (e.g., Morales et al., 2013), computer adaptive vs. paper-based testing (Puhan et al., 2007), and prior

exposure to the test (Aryadoust, 2012) shows that grouping variables in DIF studies have become increasingly diversified in more recent years.

Finally, we recorded the frequency of the different methods used to detect DIF. Because many authors utilized multiple DIF methods in a single DIF study, we counted the frequency of each DIF method that was used in each of the 110 studies. The IRT method has been used the most, 54 times, followed by MH (36 times), logistic regression (24 times), SIBTEST (14 times), Standardization (9 times), latent class method (5 times), and other methods (8 times). Among studies in which an IRT approach was used, most authors used a Rasch model or 1PL IRT. Overall, MH, logistic regression, and IRT were the most widely used methods in DIF studies in language testing. The SIBTEST, the latent class method, and the ecological model of item responding were used, albeit less frequently than other traditional DIF methods (such as MH, logistic regression, and IRT).

Research Question 2: How often do DIF Studies in Language Testing Investigate Sources of DIF?

Among the 110 studies, 45 studies did not address the source of DIF at all (i.e., Category A), 15 briefly mentioned the source of DIF (i.e., Category B), and 50 systematically examined the source of DIF (i.e., Category C). Figure 2 presents a scatter plot of the distribution of the three categories across years. A visual examination did not show any noticeable time-related patterns regarding whether researchers were more likely to systematically examine sources of DIF. Based on information in Table 2, the percentage of studies that systematically examined sources of DIF in language testing was 46.7% before 2000, 57.6% from 2000 to 2009, and 38.7% from 2010 to 2019. According to the Chi-square association test, the relationship between the three categories of DIF source detection practices and the three time periods was not

statistically significant. Therefore, there is no clear trend whether DIF studies in language testing were more likely to explain sources of DIF in more recent years.

Insert Figure 2 here

Insert Table 2 here

We also explored a few other factors for patterns in the circumstances under which researchers were more likely to systematically examine sources of DIF. For example, we looked into whether studies with data collected from certain geographic areas were more likely to do so. The Chi-square association test indicated that there was no significant relationship between the three categories of DIF source detection practices and the geographic locations of data collection. Furthermore, in terms of journals where the studies were published, the percentage of studies that systematically examined sources of DIF was highest for articles published in educational measurement journals (57.14%), followed by articles published in language testing journals (48.15%), and then articles published in other areas (33.33%). However, the Chi-square association test was not significant at .05 level, $\chi^2 = 8.328$, $df = 4$, $p = .08$. This indicates that there was no significant relationship between the three categories of DIF source detection practices and the journals where the studies were published.

Regarding the content of the test being investigated for DIF, when a test measured reading comprehension only, the study seemed to fall into Category C (systematically examined the source of DIF) more frequently. For the DIF grouping variable, studies with language as the grouping variable seemed to fall into Category C more frequently. Nonetheless, corresponding Chi-square association tests for both test content and DIF grouping were not significant. Other study characteristics, such as language of the test, native language of test-takers, did not have significant relationships with the three categories of DIF source detection practices either.

Finally, we did not conduct a Chi-square association test for DIF detection methods because many studies used multiple DIF methods. Also, with the exception of SIBTEST, which is a confirmatory DIF approach, statistical methods to detect DIF themselves usually do not directly address sources of DIF. For example, when DIF researchers who used MH, logistic regression, or IRT wanted to investigate sources of DIF, generally they would need to rely on supplementary information, such as expert content review. In contrast, when SIBTEST was the DIF detection method, sources of DIF were typically addressed in the study. With SIBTEST, researchers start by making hypotheses about a secondary dimension causing DIF based on substantive evidence and then test such hypotheses with empirical data. In addition, when explanatory variables were used in the modeling process to explain DIF, such as the ecological model of item responding, DIF sources were investigated (e.g., Zumbo et al., 2015).

Research Question 3: What Methods Have Been Used to investigate Sources of DIF?

Methods used in DIF studies before 2000. As shown in Table 2, 15 DIF application studies before 2000 were included in this review. Only seven of those studies systematically investigated sources of DIF. Among those 7 studies, Allalouf et al. (1999) was the only one that focused on investigating sources of DIF, as shown in their title “Identifying the causes of DIF in translated verbal items.” They examined the direction of DIF and asked translators to complete an item review questionnaire for each DIF items without knowledge of the DIF classification. Then, an eight-member committee reviewed the causes of DIF in each DIF item, and they found that the main reasons for DIF were changes in word difficulty, changes in item format, differences in cultural relevance, and changes in content. Similarly, Ryan and Bachman (1992) and Elder (1996) used content review to identify possible sources of DIF. Four other studies (Freedle & Kostin, 1990; Freedle & Kostin, 1997; Scheuneman & Gerritz, 1990; Schmitt, 1988)

all adopted the strategy of coding item characteristics and then running correlational or regression analyses to examine the relationship between item characteristics and DIF existence or magnitude. In summary, during the first ten years, to investigate sources of DIF, researchers tended to rely on content analysis and statistical examinations of the relationship between item characteristics and DIF status or magnitude.

Methods used in DIF studies from 2000 to 2009. During this time, DIF researchers began to expand beyond traditional methods of investigating DIF sources with expert content review, moving to more diversified and integrated methods. As shown in Table 2, among the 33 DIF studies published in this period, 19 studies systematically examined sources of DIF. Although this time was similar to the first ten years in that the majority of studies continued to draw on content experts to investigate sources of DIF, some DIF researchers started to use the multidimensional approach to DIF. Specifically, five studies (Abbott, 2007; Gierl, 2005; Jang & Roussos, 2009; Kim & Jang, 2009; Puhan et al., 2007) adopted SIBTEST, which requires both statistical and substantive judgments for identifying potential DIF items and understanding sources of DIF.

For example, Jang and Roussos (2009) used think-aloud verbal protocols to generate hypotheses about DIF items between Indo-European and non-Indo-European L2 learners. Then, they tested their DIF hypotheses using SIBTEST. Based on visual examination and review of the verbal reports and cognate types, Jang and Roussos (2009) found that DIF items with large effect sizes were associated with word translation-equivalent cognates, context-independent word meaning, and difficult distracter words. In addition, the latent class analysis approach to DIF emerged as a new DIF method (e.g., Webb et al., 2008). In contrast to standard DIF methods that are based on comparisons between an observed grouping variable, such as gender or ethnicity,

the grouping variable in Webb et al. (2008) was latent class membership. Webb et al. (2008) modelled the categorical item response options to help interpret differences in item responses associated with latent class membership. They found that DIF items between the two latent classes was primarily attributable to the use of a particular response strategy.

Methods used in DIF studies from 2010 to 2019. Among the 62 DIF studies published during this period, 24 studies systematically examined sources of DIF. As the number of DIF studies began soaring, more diversified methods were also used to investigate sources of DIF during this period. Among the 24 studies, 11 studies used the traditional approach of content review and correlational analysis between the item characteristics and DIF magnitude (Aryadoust et al., 2011; Aryadoust, 2012; Ercikan et al., 2014; Filipi, 2012; Harding, 2012; Heppt et al., 2015; Kornilov et al., 2016; Lesniewska et al., 2018; Oliveri et al., 2018; Teker et al., 2015; Wedman, 2018). Three other studies (Akcan & Kabasakal, 2019; Sandilands et al., 2013; Taylor & Lee, 2011) used SIBTEST to investigate sources of DIF, in which experts were consulted to identify the potential sources of DIF.

A few studies adopted the latent class analysis approach to DIF (Chen & Jiao, 2014; Grover & Ercikan, 2017; Oliveri et al., 2013; Oliveri et al., 2016; Zumbo et al. 2015). Despite some variation in analysis procedures, latent class DIF methods usually involved some combination of the following steps: conducting a DIF analysis with an observed grouping variable; determining the latent classes in the sample; conducting DIF with the latent groups (if there is heterogeneity) and then comparing results from the latent group DIF to results using the observed grouping variable; and examining relationships between the latent group membership and various item characteristics, person characteristics, and other contextual variables as potential sources of DIF. For example, Oliveri et al. (2013) found that differential response

patterns identified using a latent class modeling approach were more related to item performance than traditional DIF grouping variables such as gender, test language, and country. It is important to note that in Zumbo et al. (2015), latent class analysis was combined with the ecological model of item responding, which provides a comprehensive framework to explain DIF sources utilizing different layers of contextual variables.

Taking into account different contextual variables, the following DIF studies have embedded various statistical models to investigate sources of DIF. For example, Sandilos et al. (2015) used a generalized mixed logistic regression model, with persons as random effects and items as fixed effects, for the DIF analysis. They included multiple covariates that were potential sources for DIF, such as sociodemographic and language usage/exposure covariates. Similarly, Berendes et al. (2019) used a cross-classified multilevel model with random effects for person and item to examine DIF in a silent reading test between students with different home languages. Linguistic features were included in the model to explain differences in item difficulties between groups (i.e., DIF). Some of the linguistic features were significant and thus could explain the existence of DIF. Finch et al. (2016) used model-based recursive partitioning (MBRP) to examine DIF in PIRLS reading assessment across European nations, conditioning on both the reading score and the available educational resources at home. The authors found that the amount of available reading resources at home were associated with DIF for several items.

Different from the above studies where variables were controlled for in an exploratory way, Lee and Geisinger (2014) used a propensity score matching approach to investigate sources of DIF. They estimated the propensity scores based on nine contextual variables and matched the male and female groups for the DIF analysis. The authors found a noticeable reduction of DIF

items with the use of propensity score matching model, and they argued that propensity score matching was useful to hypothesize and examine potential sources of DIF in practice.

Discussion

How Are We Doing at Third Generation DIF in Language Testing?

In his article, Zumbo (2007a) described the three generations of DIF research. The first generation focused on test bias, and the second generation transitioned to the term *differential item functioning* and separated *impact* from *bias*. A number of methods to detect DIF were developed during the second generation; however, such methods (except for SIBTEST) do not address the issue of why DIF occurred. The third generation of DIF highlights the need for investigating sources of DIF. Over 13 years have passed since the publication of this article; however, it remains unknown how much DIF researchers have continued to embody the third generation praxis in language testing. Based on a thorough review of the DIF studies in language testing, we found that DIF was still a widely studied topic; however, we did not find a significant pattern that DIF researchers were more likely to address sources of DIF now than before.

We also examined other potential confounding factors. It seems that studies published in educational measurement journals and in language testing journals were more likely to investigate sources of DIF than studies published in other journals. This is understandable given that researchers who specialize in educational measurement and language testing tend to focus more attention on validity and fairness issues. In particular, researchers who publish in language testing journals are often more familiar with content issues. We found that those researchers are more likely to view DIF from a broader perspective, and to not only find DIF items but also to

identify sources of DIF. However, the Chi-square association test between the three categories of DIF source detection practices and types of journals was not significant with a p value of .08.

In addition, we observed that when a test measured reading comprehension only or when language was the DIF grouping variable, authors seemed to be more likely to systematically examine sources of DIF. However, such observed relationships did not reach statistical significance. Similarly, we did not find significant relationships between the three categories of DIF source detection practices and other study characteristics, such as the geographic location of data collection, language of the test, and native language of test-takers. It is important to note, though, that non-significant Chi-square results could also be due to small sample sizes. Therefore, those findings should be interpreted with caution given the small sample sizes in our study.

Zumbo (2007a) noted that the third generation DIF moved beyond DIF being used exclusively by large-scale testing organizations to being used across everyday research settings. Our collection of studies supported Zumbo's observations of such a transition. For example, the DIF studies in earlier years were mostly about high-stakes large-scale assessments, such as the SAT (Schmitt, 1988) and the Graduate Record Examination (GRE; Freedle & Kostin, 1990). In more recent years, DIF studies have expanded to low-stakes international tests such as PISA (e.g., Chen & Jiao, 2014; Oliveri et al., 2013, 2016), PIRLS (Stubbe, 2011) and a variety of other assessments, such as high school exit tests (Gierl, 2005), the MacArthur Communicative Development Inventory for Infants (Bruckner et al., 2007), and the Diagnostic Evaluation of Language Variation–Screening Test (Petscher et al., 2012). Also, the DIF grouping variables in earlier studies were mostly based on language, gender, or ethnicity, which is consistent with the origin of DIF in concerns about test bias and fairness in large-scale tests (Sireci & Rios, 2013). In more recent years, DIF groupings grew to include a variety of variables, including even latent

group membership. In summary, although DIF researchers in language testing have not specifically sought to assess sources of DIF more frequently in recent years than in earlier years, DIF studies in language testing has nonetheless expanded to include new DIF analysis procedures, more grouping variables, and more diversified methods for detecting DIF. Such expansions are regarded as some highlights of third generation DIF (Zumbo, 2007a; Zumbo et al., 2015).

Pros and Cons of Methods to Investigate Sources of DIF

O'Neill and McPeck (1993) highlighted three broad categories of sources of DIF: “(a) surface features or content characteristics of the question, (b) real differences in the groups’ knowledge and skills (such as those resulting from different educational experiences); or (c) the nature of the criterion used for matching” (p. 256). The methods we reviewed touch on these different causes of DIF. However, there is no gold standard to successfully detect sources of DIF. In the following, we will discuss the pros and cons of the major methods used to investigate sources of DIF based on our review of the literature.

Content Review. Review of test content has been the oldest and the most commonly used method to examine sources of DIF. For example, among the 24 studies conducted from 2010 to 2019 that systematically addressed sources of DIF, more than half still used expert content review alone or combined it with other methods. This indicates that content review has been, and still is, the primary method to investigate sources of DIF. The most common pattern we observed was that after a DIF analysis, experts were asked to review flagged DIF items to identify potential reasons for DIF occurring (e.g., Pae, 2012). However, sometimes content review was performed before DIF analysis so that the content experts did not know which items were flagged for DIF (e.g., Allalouf et al., 1999). Alternatively, other researchers relied on

student read-aloud or think-aloud protocols to identify possible sources of DIF (e.g., Jang & Roussos, 2009).

Despite the wide use of content review to understand sources of DIF, drawbacks to this method became apparent during our review. For example, content reviewers usually found it difficult to hypothesize which item(s) would show DIF and they struggled to interpret why certain items showed DIF (Gierl & Khalid, 2000). Camilli and Shepard (1994) pointed out that sometimes it was not possible for content experts to identify why a certain item showed DIF when the majority of items on a test were flagged for DIF, which Gierl et al. (1999) note is especially relevant when DIF is due to type I errors. Researchers also reported that content experts' opinions do not necessarily agree with the DIF results (Hambleton, 2006). Therefore, as Roussos and Stout (1996) concluded, "attempts at understanding the underlying causes of DIF using substantive analyses of statistically identified DIF items have, with few exceptions, met with overwhelming failure" (p. 360). Another challenge of content review is that it could be costly and time consuming to recruit and train experts (Ferne & Rupp, 2007). As such, despite the long-term and widespread use, content review is not regarded as a successful solution to identifying sources of DIF (Scott et al., 2010).

The Multidimensional DIF Framework. The Multidimensional DIF framework is typically confirmatory and relies on both statistical evidence and substantive knowledge. For example, with SIBTEST (Shealy & Stout, 1993), researchers need to identify the primary dimension, the secondary dimension, and the nuisance dimension of the test before their DIF analysis. Jang and Roussos (2009) argue that the confirmatory nature of SIBTEST yields more explanatory power than a purely exploratory approach to DIF. However, the assumption that a nuisance dimension caused DIF is a strong one and it varies based on the purpose of a given test

(Jang & Roussos, 2009; Mazor et al., 1995; McDonald, 2000). Researchers still do not fully understand the cognitive response processes underlying tests (Pepper et al., 2018), so it is very challenging to determine and identify the primary, secondary, and nuisance dimensions. Thus, even when using a fully integrated design with sophisticated procedures, Jang and Roussos (2009) admit that deciding whether DIF items or bundles measure an unintended nuisance dimension or are part of the intended multidimensional components requires a “great deal of well-reasoned arguments and additional evidence” (p. 244). Moreover, Zumbo et al. (2015) argue that the multidimensional approach to DIF only looks at item characteristics and cognitive processes, which limits information about sources of DIF. As such, compared to methods such as MH, logistic regression, or IRT, use of the multidimensional DIF framework has been limited.

Third Generation DIF Methods. Different from the second generation of DIF, which was characterized by both statistical evidence and substantive evidence to investigate sources of DIF, third generation DIF methods are informed by an explanation-focused view of validity (Zumbo, 2007a; Zumbo et al., 2015). With third generation approaches, researchers go beyond the standard assumption that DIF is caused by item characteristics to include a broader range of learning contexts, such as a student’s background characteristics or instructional variables.

For example, Zumbo et al. (2015) proposed that third generation DIF methods can be applied within the broader ecological model of item responding. To demonstrate this model, Zumbo et al. (2015) used a latent class approach to DIF to analyze the PISA reading dataset. They included 19 explanatory variables to explain DIF and four of them emerged as statistically significant. Woitschach et al. (2019) used a multilevel modeling approach to explain why gender DIF occurred on the TERCE science test. After examining two student-, two school-, and two county-level characteristics as the explanatory variables, the authors concluded that the main

source of gender DIF was related to the human developmental level of the 15 participating countries. Both studies used an exploratory approach with a large-scale international dataset from multiple countries. However, given that the ecological model is relatively new, the application of this model in language testing is still not common.

Zumbo et al. (2015) proposed five layers of explanatory variables and indicated that more layers of variables could be included. Implicit in successfully applying the ecological model to explain DIF is that researchers not only have a comprehensive understanding of the test takers' ecological environment, but also that they have access to those ecological variables, which is not always the case for smaller datasets with little background information. For smaller datasets with limited background information, Zumbo et al. (2015) suggested that researchers could use a theory-testing approach to select explanatory predictors instead of the modeling approach generally taken with international assessment data.

The ecological model can also be combined with different statistical methods to DIF. For example, Zumbo et al. (2015) used the ecological model in conjunction with a latent class analysis approach. In traditional DIF approaches, the grouping variable is usually an observed trait with the assumption that everyone in each group is homogeneous. However, the homogeneity assumption is not always met, especially for datasets involving participants from multiple countries in large-scale international tests. Although latent class analysis addresses homogeneity issues, various researchers (e.g., Chen & Jiao, 2014; Oliveri et al., 2016; Zumbo et al., 2015) have nonetheless pointed out some general challenges and limitations with the latent class approach to DIF as a statistical method. For example, it is not an easy task to derive meaningful latent classes. Depending on the sample size, the specific analysis procedure, and the model fit criteria, latent classes may be over- or under-extracted (Cho et al., 2012; Li et al., 2009).

These limitations, however, are specific to latent class analysis methods, and do not reflect any inherent limitations of the ecological model of item responding.

Conclusion, Limitation, and Implications

DIF analysis is an important method used to address test fairness, which is a significant topic in educational measurement (AERA, APA, & NCME, 2014). The present study provides a review of the status and trend of DIF research in language testing. In particular, we provide a critical review of the methods used to investigate sources of DIF in language testing. A primary limitation of this study is that we only included peer reviewed journal articles, while book chapters, research reports, conference presentations, and other unpublished work were excluded. Also, we only considered papers written in English. Therefore, our review may not fully represent the DIF literature in language testing.

We found that DIF researchers did not assess sources of DIF more frequently in recent years than in earlier years; nevertheless, DIF research in language testing has expanded with new DIF analysis procedures, more grouping variables, and more diversified methods for investigating sources of DIF. In the early years of DIF research, methods to identify sources of DIF relied heavily on expert analysis of item content as well as correlational analysis between item characteristics and DIF presence or magnitude. Later on, more statistically sophisticated procedures were proposed to specifically address sources of DIF, such as SIBTEST and latent class analysis. More recently, Zumbo et al. (2015) introduced the ecological model of item responding, in which different layers of explanatory variables are used to explain DIF.

Despite such statistical advances and laudable efforts, however, our review shows that substantiating sources of DIF remains the most challenging task in DIF research (Camilli &

Shepard, 1994; Douglas et al., 1996; Gierl, 2005). Among the studies we reviewed, many authors appeared to be tentative when making interpretations about what caused DIF in their study, even after a systematic investigation with well-designed procedures (e.g., Chen & Jiao, 2014; Jang & Roussos, 2009). One reason for this hesitation may be that it is difficult to determine the DIF status of an item. The existence of DIF in an item depends on the particular DIF method, the specific criteria for detecting DIF, sample size, power, type I error, etc. Consistent with prior literature (Hambleton, 2006), we observed that researchers would often use multiple methods for DIF detection and would see inconsistent DIF results across those multiple methods. Furthermore, DIF studies are usually observational and thus permit only inferences about the relationships between item characteristics and DIF (Schmitt et al., 1993). Therefore, DIF researchers were understandably reluctant to claim that they have indeed found the actual sources of the DIF.

Although it is not practical to conduct experimental design studies to identify sources of DIF for most tests, we believe that researchers can still attempt to explain DIF by accumulating evidence collected from multiple approaches and multiple studies. While DIF is a statistical procedure to address test fairness issues, DIF has also been regarded as part of the validity framework. Informed by an explanation-focused view of validity (Zumbo, 2007b; Zumbo, 2009), the ecological model of item responding aims to explain DIF and to further link DIF research to the overall test validity. From this perspective, just as test validation is an ongoing process and not a one-time effort (Messick, 1995), so too is assessing and explaining DIF. As Kunnan (2007) noted, DIF studies are not “one-off,” but instead they contribute to a larger picture of validity evidence. That is, understanding sources of DIF is a cumulative process, and multiple methods need to be adopted to deepen the understanding from different perspectives. In addition to

statistical and psychometric techniques to investigate sources of DIF, from the perspective of the ecological model of item responding (Zumbo et al., 2015), we encourage DIF researchers to emphasize acquiring theoretical and substantive knowledge of test content as well as characteristics of test-takers and their learning environment.

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Table 1*Characteristics of the Collected DIF Studies in Language Testing*

Characteristics	Subcategory	Frequency	Percentage
Time period	Before 2000	15	13.6%
	2000 to 2009	33	30.0%
	2009 to 2019	62	56.4%
Geographic location of data collection	U.S. and Canada	49	44.5%
	Asia	25	22.7%
	Europe	13	11.8%
	Australia & New Zealand	6	5.5%
	Other places	4	3.6%
	Multiple locations	13	11.8%
Journal where the study is published	Journals focused on language testing	27	24.5%
	Journals focused on educational measurement	41	37.3%
	Other journals	42	38.2%
Content of the test	Reading only	41	37.3%
	Vocabulary only	12	10.9%
	Listening only	7	6.4%
	Other skills (oral, writing, pragmatics)	4	3.6%
	Multiple skills	46	41.8%
Language of the test	English	83	62.9%
	Spanish	8	6.1%
	French	8	6.1%
	Russian	6	4.5%
	German	5	3.8%
	Chinese	4	3.0%
	Arabic	4	3.0%
	Other languages	14	10.6%
Native language of test-takers (for the 83 studies with tests in English)	English only	28	33.7%
	English and another language	15	18.1%
	Farsi only	5	6.0%
	Korean only	5	6.0%
	Japanese only	4	4.8%
	Turkish only	3	3.6%
	Spanish, Finish, French, or Swedish only	5	6.0%
	Multiple native language groups	18	21.7%
Whether DIF items have been found in the study	Found DIF items	103	93.6%
	Did not find DIF items	7	6.4%

DIF grouping variable	Gender	41	28.5%
	Language (native language, home language, etc.)	37	25.7%
	Ethnicity	23	16.0%
	Age	9	6.3%
	Social economic status	5	3.5%
	Academic discipline	5	3.5%
	Rural vs. urban	3	2.1%
	Other	21	14.6%
Method for DIF detection	IRT	54	36.0%
	MH	36	24.0%
	Logistic regression	24	16.0%
	SIBTEST	14	9.3%
	Standardization	9	6.0%
	Latent class analysis	5	3.3%
	Other	8	5.3%

Table 2*Distribution of Studies across DIF Source Detection Practices*

Characteristics	Subcategory	Category A	Category B	Category C	Chi-square association test
Time period	Before 2000	6	2	7	$\chi^2 = 6.389$, df = 4, $p = .172$
	2000 to 2009	8	6	19	
	2009 to 2019	31	7	24	
Geographic location of data collection	U.S. and Canada	16	7	26	$\chi^2 = 5.658$, df = 6, $p = .463^a$
	Asia	15	2	8	
	Europe	5	1	7	
	Australia & New Zealand	0	3	3	
	Other	4	0	0	
	Multiple locations	5	2	6	
Journal where the study was published	Journals focused on language testing	9	5	13	$\chi^2 = 8.328$, df = 4, $p = .080$
	Journals focused on educational measurement	12	6	23	
	Other journals	24	4	14	
Content of the test	Reading only	13	5	23	$\chi^2 = 6.276$, df = 4, $p = .179^b$
	Vocabulary only	8	2	2	
	Listening only	2	0	5	
	Other skills (oral, writing, pragmatics)	3	1	0	
	Multiple skills	19	7	20	
Language of the test	English	33	14	36	$\chi^2 = 2.433$, df = 3, $p = .487^c$
	Spanish	5	0	3	
	French	2	0	6	
	Russian	0	0	6	
	German	1	1	3	
	Chinese	2	0	2	
	Arabic	2	0	2	
	Other	6	0	8	
Native language of test-takers (for the 83 studies with test in English language)	English only	10	5	13	$\chi^2 = .540$, df = 4, $p = .970^d$
	English and another language	5	2	8	
	Farsi only	5	0	0	
	Korean only	1	2	2	
	Japanese only	4	0	0	
	Turkish only	1	0	2	
	Spanish, Finish, French, or Swedish only	2	2	1	
	Multiple native language	5	3	10	

	groups			
DIF grouping variable	Gender	20	5	16
	Language	9	5	23
	Ethnicity	8	5	10
	Age	5	1	3
	Social economic status	1	0	4
	Academic discipline	4	1	0
	Rural vs. urban	2	0	1
	Other	9	1	11

$\chi^2 = 9.192$,
 $df = 8$, p
 $= .326^e$

Note. Category A = did not mention the source of DIF, Category B = briefly mentioned the source of DIF, Category C = systematically examined the source of DIF

- a. "Australia & New Zealand" and "Other" were not included in the Chi-square test.
- b. "Listening only" and "Other skills" were not included in the Chi-square test.
- c. "Russian," "German," "Chinese," "Arabic" and all cells in Category B were not included in the Chi-square test.
- d. "Farsi only," "Korean only," "Japanese only," "Turkish only," and "Spanish, Finish, French, or Swedish only" were not included in the Chi-square test.
- e. "Social economic status," "Academic discipline" and "Rural vs. urban" were not included in the Chi-square test.

Figure 1

Distribution of DIF Studies across Years

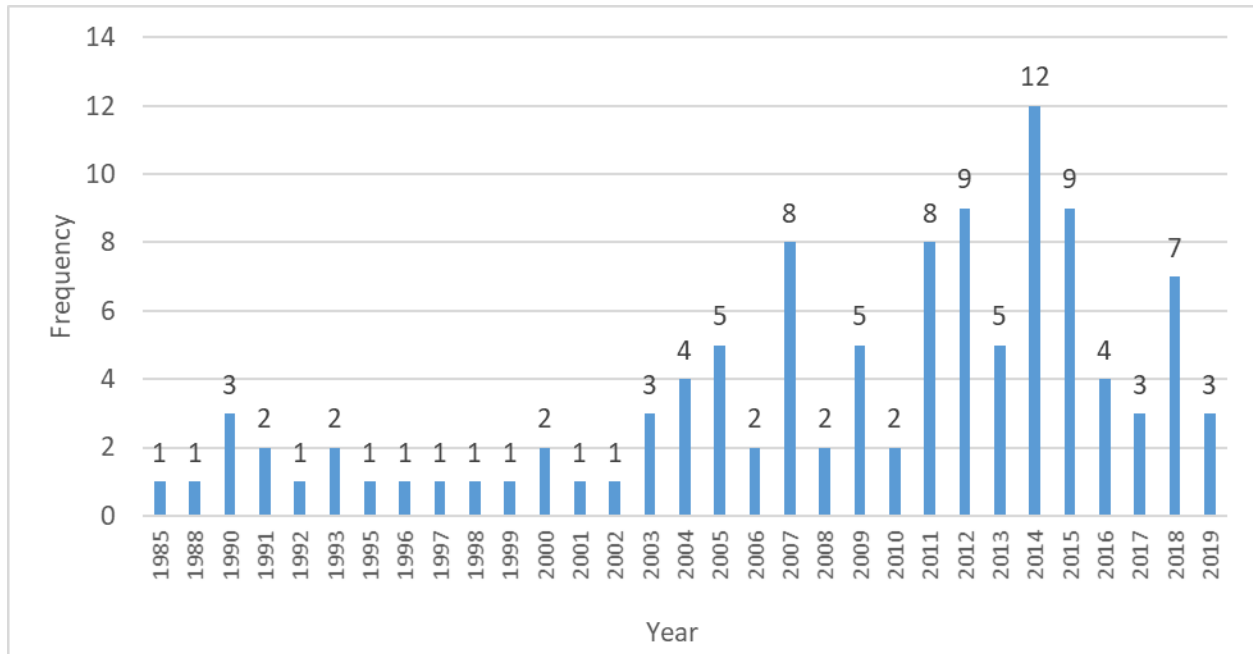
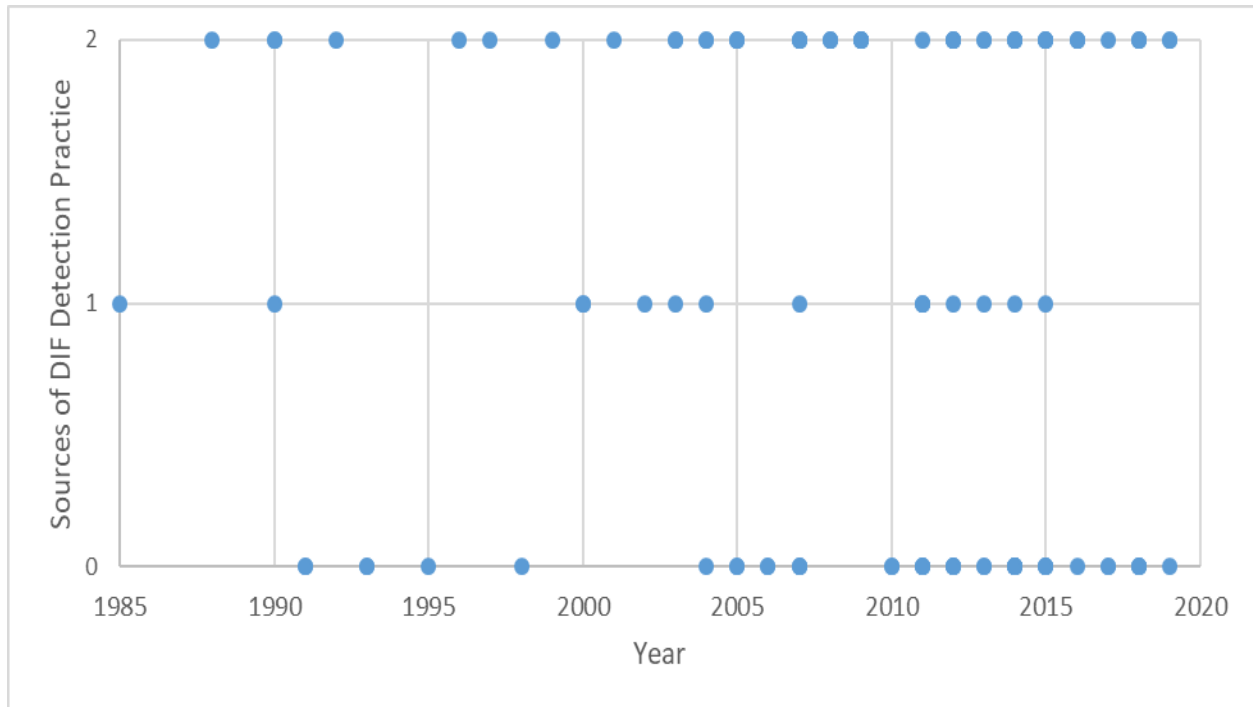


Figure 2

DIF Studies That Have Systematically Examined Sources of DIF over Years



Note. 0 = Category A (did not mention the source of DIF). 1= Category B (briefly mentioned the source of DIF). 2 = Category C (systematically examined the source of DIF)

Acknowledgement: The authors are indebted to Antony Kunnan and two anonymous reviewers for their valuable suggestions.

Conflict of interest statement: We have no potential conflict of interest.